Rahul Choudhary

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Senior Full Stack Engineer

ABOUT ME

- Full Stack Developer with 4+ years of experience in MERN stack, Next.js, and modern JavaScript (ES2017+), specializing in building scalable, high-performance applications.
- Strong expertise in software architecture and design patterns (SOLID) with a deep understanding of functional programming concepts to develop maintainable and efficient code.
- Proven leadership experience, leading development teams, driving best practices, mentoring engineers, and ensuring successful project execution from planning to deployment.
- Hands-on experience with deployment and DevOps, managing cloud-based infrastructure, optimizing CI/CD pipelines, and ensuring high availability and reliability of applications.
- Experienced in AI-driven automation, leveraging n8n AI to create smart agents that enhance workflows and improve application efficiency.
- Passionate about Large Language Models (LLMs), exploring emerging technologies, and optimizing existing systems for better performance, scalability, and user experience.

TECHNICAL SKILLS

Programming Languages & Core Technologies

- **Languages**: JavaScript (ES2017+), TypeScript, HTML5, CSS3, SQL
- **Backend**: Node.js, Express.js, RESTful APIs, GraphQL (Apollo Client/Server), WebSockets
- **Frontend**: React.js, Next.js, Redux, TanStack React Query
- **Databases**: MongoDB, PostgreSQL, Redis, Firebase
- **ORMs**: TypeORM, Sequelize, Prisma

Cloud & DevOps

- **Cloud Platforms**: AWS (EC2, S3, Lambda,Route53), Firebase
- **Containerization**: Docker, Kubernetes
- **Deployment**: Serverless Framework, CI/CD Pipelines
- **AI Deployment**: Containerized AI models, Serverless AI functions, Edge AI deployment

UI/UX & Styling

- **Frameworks**: Tailwind CSS, Material-UI (MUI) Shadcn
- **AI-Powered UI Tools**: Integration with AI-assisted design tools
- **Responsive Design**: Mobile-first approach, Progressive Web Apps

Architecture & Best Practices

- **Architecture**: Microservices, Event-driven architecture, AI-first application design
- **Security**: Authentication & Authorization (JWT, OAuth), OWASP Security Principles
- **Testing**: Jest,Enzyme, React Testing Library,
- **AI-Specific**: Prompt engineering, Model fine-tuning, RAG implementation , ChatBot

API & Microservices: RESTful APIs, GraphQL (Apollo Client/Server), WebSockets, Microservices Architecture

AI & Generative AI Technologies

AI & LLMs: ChatGPT 3.5 Turbo, ChatGPT 4.0, ChatGPT 4.0 Mini, ClaudeAI, Grok, DeepSeek, Google Gemini (1.0 & 2.0), Bolt.new, V0 by Vercel , Perplexcity, freePick , Creatomate , Elavan Labs

AI Agent Tools: n8n (Automation & Workflow Optimization)

AI Integration Tools:

- MCP (Model Context Protocol) Server - TypeScript SDK,
- Cursor AI with RAG (Retrieval-Augmented Generation)
- Vector databases (Pinecone , Supabase) for semantic search
- Embedding models for document processing

MY STRONG SUITS

- Proven track record of delivering high-quality software applications that meet customer needs and provide value to the business.
- Extensive experience working with modern JavaScript (ES2017+) and a willingness to learn new technologies, demonstrating the ability to stay current with emerging trends in the industry.
- Proficient in building scalable and reliable web applications using Node.js, PostgreSQL, and React.js, ensuring high-performance and maintainability.
- Skilled in applying SOLID principles to real-world applications, ensuring clean, maintainable, and testable code.
- Passionate about writing clean code and building well-tested software, emphasizing a commitment to delivering high-quality products.
- Built various applications from scratch using ReactJS and NodeJS, demonstrating expertise in full-stack development.
- UI development experience with Material-UI, Ant Design, Tailwind CSS, and Bootstrap, ensuring modern and responsive user interfaces.
- Expertise in microservices architecture and message brokers, including NATS, RabbitMQ, and BullQueue, facilitating efficient inter-service communication.
- Experienced in designing and implementing workflow automation using n8n, optimizing business processes, and reducing manual effort.
- Good experience in hosting private n8n instances, ensuring secure and scalable workflow automation for enterprise environments.
- Skilled in integrating Large Language Models (LLMs) like ChatGPT, ClaudeAI, and Gemini to enhance applications with AI-driven capabilities.

- Developed AI-powered smart agents using n8n, improving application efficiency, automation, and decision-making.
- Successfully set up the DeepSeek model on Amazon Bedrock, leveraging cloud-based AI services for advanced natural language processing.

WORK EXPERIENCE

Full Stack Developer | MenteStack | Aug 2024 - Present

- Collaborated with product managers and UX designers to develop and deliver high-quality web applications that meet customer needs.
- Utilized SOLID and GRASP principles to design and implement maintainable and testable code.
- Developed responsive and accessible user interfaces using React and Redux, ensuring a seamless user experience.

🔍 **AI, GenAI, and Workflow Automation (9 + months experience)**

- Built end-to-end **AI-powered n8n workflows** using **smart agents** and **OpenAI, ClaudeAI, Google Gemini**, and **DeepSeek** models via **Amazon Bedrock**.
- Integrated **LLM-based agents** with Telegram,Slack and other interfaces to automate intelligent decision-making and reporting.
- Designed and implemented **RAG (Retrieval-Augmented Generation)** pipelines using **vector databases** like **Pinecone,Supabase**, and **Qdrant**.
- Implemented **semantic search**, **vector embeddings**, and **custom prompt chains** for high-accuracy information retrieval.
- Used **LangChain** to build simple **LLM chains**, leveraging **LLMChain**, **PromptTemplate**, and **Tool usage** for agent-driven architectures.
- Automated document analysis and report generation via **PDF/Image input extraction**, image-to-text agents,voice-to-text and health data processing.
- Employed **Google Apps Script** to connect spreadsheet data with AI agents, enabling dynamic data workflows.
- Set up and manage **secure private n8n instances** for enterprise-grade workflow orchestration.
- Worked with **MCP cloud servers** to deploy and scale AI-native applications efficiently.
- Created **chatbots with memory and context awareness** that interact with structured/unstructured data.
- Used summarization chains and context-aware prompts to deliver real-time responses with **RAG pipelines.**
- Combined image analysis, MIME type detection, LLM summarization, and health metric analysis into cohesive agent workflows.

Full Stack Developer | Encred Analytics Odisha | Oct 2023 - Aug 2024 | On Site

- Developed and maintained web applications using modern JavaScript frameworks and libraries.
- Collaborated with cross-functional teams to identify and resolve technical issues and improve the application's performance.
- Utilized object-oriented analysis and design principles to create maintainable and testable code.

- Followed best practices in software development, including continuous integration and delivery, code reviews, and automated testing.
- Implemented responsive and accessible user interfaces using React and Redux.

Full Stack Developer | Santophy Infotech Indore | Jan 2021 - Oct 2023 | On Site

- Built and maintained scalable web applications using **React.js**, **Redux**, **Node.js**, and **Express.js**, improving load time and user experience by 30%.
- Integrated **RESTful APIs** and optimized backend logic, reducing API response times by 40%.
- Utilized version control tools (**Git**, **GitHub**) and Agile methodologies to manage project lifecycle and deliver sprints on time.
- Created dynamic and modular UI components with **React Hooks**, **Context API**, and **Redux Toolkit** for efficient state management.
- Led a team of 4–6 developers, overseeing project planning, task delegation, and code quality assurance through structured **code reviews** and **pair programming**.

PROJECTS

Project 1: AI-Powered Health Assistant Platform

Industry: Healthcare
Role: Full Stack AI Engineer

An AI-driven health assistant platform that empowers users to make informed dietary and lifestyle choices. The platform allows users to ask health-related queries via both a **Telegram bot** and a **web app**. Users can upload health reports (PDFs/images), which are analyzed by AI agents to generate **Risk Insight Outputs (RIOs)** based on health metrics. The system also connects with **Oura Ring data**, enabling personalized wellness analysis. A robust memory system ensures context-aware responses, while an AI chatbot guides users with advice on food choices, symptoms, and overall health.

Technologies Used: n8n AI Agent Tool, Pinecone, RAG (Retrieval-Augmented Generation), Chatgpt 4o mini LLM,Google App Script,

Responsibilities:

- Built and orchestrated **modular AI workflows** in **n8n**, connecting multiple agents for report analysis, image processing, summarization, and contextual response generation.
- Enabled **multi-modal input handling**, including Telegram bot uploads and web app queries for real-time analysis.
- Integrated **RAG pipelines using FAISS**, ensuring grounded, hallucination-free responses by fetching semantically matched context before generating LLM answers.
- Connected and processed **Oura Ring data**, summarizing sleep readiness and activity for health assessments.
- Implemented **image-to-text (OCR)** pipeline with MIME type detection for extracting key info from uploaded health files.
- Designed a **robust memory system** to store and recall user inputs, summaries, and AI responses for a personalized experience.

- Built a responsive UI in **React + TailwindCSS** with clean component documentation via **React Storybook**.
- Utilized **React Query + Context API** for efficient state and async data handling.
- Set up a full **CI/CD pipeline** and deployed the system on an **Amazon EC2 instance** for secure and scalable access.

Project 2: AI-Driven FedEx Shipping Assistant

Industry: Logistics & Shipping
Role: Full Stack AI Engineer

Built an AI-powered shipping assistant integrated with a Telegram bot to automate the entire shipping lifecycle, from collecting shipping data to generating FedEx rate/shipping details and storing everything in Google Sheets. The system interacts conversationally with users to collect shipping inputs (including images and dimensions via iPhone), validates and enriches the data using agents, and ensures complete and accurate record-keeping.

Technologies Used: n8n AI Agent Tool, RAG (Retrieval-Augmented Generation), Chatgpt 4o mini LLM,Google App Script, Supabase

Responsibilities:

Designed and implemented a complete AI-agent workflow in **n8n** that:

- Accepts shipping details via **Telegram** (supports both text and image-based input).
- Authenticates with **FedEx API** to fetch real-time rate and shipping data.
- Parses package dimensions, including from **iPhone-captured images** using AI.
- Automatically validates required fields and requests missing data from users via chat.
- Merges and post-processes multiple API responses to extract structured shipping details.
- Stores clean, validated shipping data into **Google Sheets** for further use.

Integrated multiple **AI agents via OpenRouter** to handle:

- **Payload generation** based on user inputs and context.
- **Validation** of user-provided data and API responses.
- **Data extraction** from unstructured sources (e.g., text/image).
- **Structured aggregation** of all workflow outputs for consistent recordkeeping.
- Implemented modular data processing with **Tools Agent** and **Structured Output Parser** for maintainable logic.
- Designed an acknowledgment flow to notify users of the shipping progress.
- Enabled scalable, hands-free shipping automation through conversational interaction.

Project 3: Snippit (CRM)

Industry: SaaS / Customer Relationship Management
Role: Full Stack Developer

Snippit was a modern Customer Relationship Management (CRM) platform built to transform how businesses capture leads, manage pipelines, and maintain client communication. Designed for scalability and real-time collaboration, it offered customizable dashboards,

automated workflows, and seamless integrations to boost sales productivity and customer engagement.

Responsibilities:

- **Responsive UI Development:** Designed and developed mobile-first, responsive user interfaces using React and PrimeReact, ensuring smooth cross-device usability.
- **API Integration:** Created and consumed secure RESTful APIs with Express.js and integrated third-party tools like email services, calendar syncing, and CRM enrichment APIs.
- **Full CRUD Modules:** Built reusable components and modules for leads, contacts, tasks, and pipelines with complete create-read-update-delete functionality.
- **Data Management & Mapping:** Handled complex entity relationships using **TypeORM**, including user roles, lead stages, and activity logs with custom query builders.
- **Authentication & Authorization:** Implemented JWT-based user authentication and role-based access control across the app.
- **State Management:** Used **Redux Toolkit** to manage complex UI and form state, enabling consistent user experience across components.
- **Performance Optimization:** Applied lazy loading, code-splitting, and API caching strategies to improve app performance.
- **CI/CD Pipelines:** Set up GitHub Actions for automated testing, builds, and deployments to AWS Amplify.
- **Cloud Integration:** Managed static asset hosting on **AWS S3**, connected server to PostgreSQL instances, and optimized build deployments via Amplify.
- **Collaboration & Agile Development:** Participated in sprint planning, peer reviews, and daily scrums. Maintained Git branches and pull requests for clean DevOps flow.
- **Testing & Debugging:** Wrote component-level and integration tests using Jest and React Testing Library to ensure reliability and maintainability.